

Hyper tuning parameter assignment

Dataset filename: **50 Startups.csv**

R2_score comparison for different algorithms like Multiple Linear Algorithm, Support Vector Machine, Decision Tree by creating multiple models with different hyper parameters are below:

1) Multiple Linear Algorithm:

R2value is **0.9358680970046241**

2) SVM (Regression) Algorithm:

S. No	Hyper parameter	LINEAR (r value)	RBF (NON LINEAR) (r value)	POLY (r value)	SIGMOID (r value)
1	C10	-0.039644946	-0.05680	-0.053667	-0.0547195
2	C100	0.10646819	-0.0507260	-0.019802	-0.0304535
3	C500	0.5928977	-0.0243233	0.1146848	0.0705721
4	C1000	0.7802839882	0.00676834	0.266163	0.18506861
5	C3000	0.89567446	0.12322756	0.63700642	0.591363
6	C15000	0.930124939	0.475584748	0.763866	0.873377

the SVM regression best R2value in (Linear and hyper parameter C=15000) is = **0.930124939**

3) Decision Tree Algorithm:

S.NO	CRITERION	MAX FEATURES	SPLITER	R VALUE
1	squared_error	sqrt	best	0.7450432
2	squared_error	sqrt	random	0.0122444
3	squared_error	Log2	best	0.8038820
4	squared_error	Log2	random	0.8373856
5	squared_error	none	best	0.9263877
6	squared_error	none	random	0.9289554
7	friedman_mse	sqrt	best	0.8881054
8	friedman_mse	sqrt	random	0.5564338
9	friedman_mse	Log2	best	0.6592685
10	friedman_mse	Log2	random	0.5130277
11	friedman_mse	none	best	0.9021996
12	friedman_mse	none	random	0.7141621
13	absolute_error	sqrt	best	0.7804672
14	absolute_error	sqrt	random	-0.7486643
15	absolute_error	Log2	best	0.7098025
16	absolute_error	Log2	random	0.4387119
17	absolute_error	none	best	0.9490915
18	absolute_error	none	random	0.6790464

the Decision Tree best R2value in (criterion= absolute_error, Max features= None, splitter=best) is = **0. 9490915**. Decision Tree Algorithm give the best R2 score, so it the best model